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AUTOMATED DECOUPLED CHARGER FOR ELECTRIFIED VEHICLE

Abstract

A vehicle charging system includes a charging base, a cable arm, a charging cable, a controller and a vehicle charge connector. The controller is configured to determine whether a decoupling condition has been satisfied and communicate a decoupling signal in response thereto. The vehicle charge connector is configured to be electrically coupled to a vehicle charging port of the electrified vehicle during a charging event. The vehicle charge connector includes a charger body and an actuator. The charger body has a male extension portion extending therefrom. The actuator is movably disposed on the charger body and is configured to move from a retracted position to a deployed position upon receipt of the decoupling signal. Movement of the actuator to the deployed position causes a forcing surface on the actuator to push against a surface on the electrified vehicle and urge the vehicle charge connector away from the vehicle charging port.

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Background/Summary

FIELD

[0001] The present application relates generally to electrified vehicles and, more particularly, to an electrified vehicle charging system having a charging cable coupling assembly that automatically detaches from the electrified vehicle.

BACKGROUND

[0002] An electrified vehicle (hybrid electric, plug-in hybrid electric, range-extended electric, battery electric, etc.) includes at least one battery system and at least one electric motor. Typically, the electrified vehicle could include a high voltage battery system and a low voltage (e.g., 12 volt) battery system. In such a configuration, the high voltage battery system is utilized to power at least one electric motor configured on the vehicle and to recharge the low voltage battery system via a direct current to direct current (DC-DC) convertor. Electrified vehicles require an electrical charging cord, such as a Type 2/Level 2 portable charger, that electrically couples between a power source and the vehicle battery. Typically, when the electrified vehicle has reached a desired charge status, a user manually decouples an electrical charging connector associated with the electrical charging cord from the vehicle and returns it to the charging system. Accordingly, while such electrified vehicle charging connections do work for their intended purpose, there exists an opportunity for improvement in the relevant art.

SUMMARY

[0003] In accordance with one example aspect of the invention, a vehicle charging system for an electrified vehicle is provided. The vehicle charging system includes a charging base, a cable arm, a charging cable, a controller and a vehicle charge connector. The charging base houses electrical charging components. The cable arm is movably coupled to the charging base. The charging cable is configured to communicate current from the electrical charging components to the electrified vehicle. The controller is configured to determine whether a decoupling condition has been satisfied and communicate a decoupling signal in response thereto. The vehicle charge connector extends from the charging cable and is configured to be electrically coupled to a vehicle charging port of the electrified vehicle during a charging event. The vehicle charge connector includes a charger body and an actuator. The charger body has a male extension portion extending therefrom. The actuator is movably disposed on the charger body and is configured to move from a retracted position to a deployed position upon receipt of the decoupling signal. Movement of the actuator to the deployed position causes a forcing surface on the actuator to push against a surface on the electrified vehicle and urge the vehicle charge connector away from the vehicle charging port.

[0004] In addition to the foregoing a motor is disposed on the charging base and mechanically coupled to the cable arm. The motor configured to move the cable arm from an outward charging position to a retracted position.

[0005] In addition to the foregoing, the controller communicates a signal to the motor to move the cable arm to the retracted position subsequent to communicating the decoupling signal.

[0006] In addition to the foregoing, the charging base further comprises a magnet, wherein the magnet is configured to magnetically couple the vehicle charge connector thereat.

[0007] In addition to the foregoing, the controller determines that a decoupling condition has been satisfied based on a vehicle charge is complete.

[0008] In addition to the foregoing, the controller determines that a decoupling condition has been satisfied based on an unsafe condition being detected

[0009] In addition to the foregoing, the controller determines that a decoupling condition has been

satisfied based on an indication that the electrified vehicle has shifted out of park.
[0010] In addition to the forgoing, the charger body includes a track assembly, wherein the actuator translates along the track assembly from the retracted position to the deployed position.
[0011] In addition to the forgoing, the vehicle charging port further defines a female receiving portion that receives the male extension portion of the charger body during the charging event.
[0012] According to another example aspect of the invention, a method for automatically decoupling a vehicle charge connector from an electrified vehicle is provided. The method includes determining, at a controller provided in a charging base that houses electrical charging components, that a charge condition is enabled; determining, at the controller, whether a decoupling condition has been satisfied; and communicating, based on a decoupling condition being satisfied, a decoupling signal to an actuator on the vehicle charge connector, the actuator being caused to move from a retracted position to a deployed position upon receipt of the decoupling signal, wherein movement of the actuator to the deployed position causes a forcing surface on the actuator to push against a surface on the electrified vehicle and urge the vehicle charge connector away from the vehicle charging port.
[0013] In addition to the forgoing, the method further includes communicating a signal to a motor disposed in the charging base, the motor mechanically coupled to a cable arm, whereby the motor moves the cable arm from an outward charging position to a retracted position.
[0014] In addition to the forgoing, the signal is communicated to the motor subsequent to communicating the decoupling signal.
[0015] In addition to the forgoing, the method includes magnetically coupling the vehicle charge connector to a magnet disposed on the charging base subsequent to the cable arm moving to the retracted position.
[0016] In addition to the forgoing, the decoupling condition is satisfied based on an indication that a vehicle charge is complete.
[0017] In addition to the forgoing, the decoupling condition is satisfied based on an indication that an unsafe condition is detected.
[0018] In addition to the forgoing, the decoupling condition is satisfied based on an indication that the electrified vehicle has shifted out of park.
[0019] In addition to the forgoing, moving the actuator from the retracted position to the deployed position comprises translating the actuator along a track.
[0020] Further areas of applicability of the teachings of the present disclosure will become apparent from the detailed description, claims and the drawings provided hereinafter, wherein like reference numerals refer to like features throughout the several views of the drawings. It should be understood that the detailed description, including disclosed embodiments and drawings references therein, are merely exemplary in nature intended for purposes of illustration only and are not intended to limit the scope of the present disclosure, its application or uses. Thus, variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0021] FIG. **1** is a perspective view of an electrified vehicle charging system having a charging cable coupling assembly that automatically detaches from the vehicle according to the principles of the present disclosure;
[0022] FIG. **2**A is a front perspective view of a vehicle charge connector that selectively couples to an electrified vehicle, the vehicle charge connector shown in a coupling position with an actuator in a retracted or at-rest position according to an example of the present disclosure;

[0023] FIG. **2**B is a front perspective view of the vehicle charge connector of FIG. **2**A shown in a decoupling position with the actuator in a deployed position according to an example of the present disclosure;
[0024] FIG. **3** is a front perspective view of the electrified vehicle charging system shown with a user initially coupling the vehicle charge connector to an electrified vehicle to initiate a charging event;
[0025] FIG. **4**A is a side view of the vehicle charge connector of FIG. **2**A shown in a connected position with the electrified vehicle;
[0026] FIG. **4**B is a side view of the vehicle charge connector of FIG. **2**B shown in a decoupled position subsequent to the actuator moving to the deployed location;
[0027] FIG. **5** is a front perspective view of the electrified vehicle charging system shown subsequent to a decoupling signal communicated to the actuator to unplug the vehicle charge connector from the electrified vehicle, wherein an arm automatically returns the charging cable toward the charging base;
[0028] FIG. **6** is a front perspective view of the electrified vehicle charging system shown with the vehicle charge connector magnetically attached to a magnet on the charging base; and
[0029] FIG. **7** is a logic flow chart illustrating steps for automatically detaching the vehicle charge connector from an electrified vehicle subsequent to receiving a decoupling signal and returning the vehicle charge connector to the charging base according to examples of the present disclosure.

DESCRIPTION

[0030] As previously discussed, there exists an opportunity for improvement in the art of charging electrified vehicles. For example, electrified vehicles are generally required to electrically connect a cable between the vehicle and a power source for charging the high-voltage battery of the electric vehicle (A/C charging). Typically, a user physically takes the vehicle charge connector from a vehicle charging station and couples the vehicle charge connector to their vehicle. When the desired charge is completed, the user physically decouples the vehicle charge connector from their vehicle and returns it to the vehicle charging station. The act of manipulating the vehicle charge connector and associated electrical cable that extends between the vehicle charge connector and the vehicle charging station can be cumbersome. In this regard, many electrical cables are heavy and awkward to maneuver.
[0031] The present disclosure provides a vehicle charging system that incorporates a charging cable coupling assembly that automatically detaches from the vehicle. Upon indication of a decoupling signal, the vehicle charge connector automatically decouples from the vehicle. Furthermore, a cable arm that orients the charging cable retracts from a charging position to a stowed position bringing the charging cable and the vehicle charge connector back toward the charging base and generally away from the vehicle. A magnet on the charging base magnetically attracts the vehicle charge connector to an attached position on the charging base. The automatic decoupling alleviates the need for the vehicle user to manually disconnect the vehicle charge connector from the vehicle and return it to the charging base. As described herein, the decoupling signal that triggers the decoupling can result from a variety of conditions such as, but not limited to, an indication that charging has reached the desired limit, an unsafe condition, unauthorized use of the charging system, emergency use of a connected vehicle.
[0032] Referring now to FIG. **1**, vehicle charging system constructed in accordance to examples of the present disclosure is shown and generally identified at reference numeral **10**. The vehicle charging system **10** generally includes a charging base **12**, a cable arm **14** and a charging cable coupling assembly **20**. The charging cable coupling assembly **20** generally comprises a charging cable **40** and a vehicle charge connector **42**. The charging cable coupling assembly **20** is configured to communicate current from the charging base **12** and into an electrified vehicle during a charging event.
[0033] The charging base **12** generally houses electrical charging components **22** within a housing

24. A user interface panel **30** and a magnet **28** are arranged on the housing **24** of the charging base **12**. A motor **34** drives a shaft **36** operably coupled to the cable arm **14**. As will be described in greater detail herein, rotation of the shaft **36** causes the cable arm **14** to rotate around an axis **38**. The user interface panel **30** can receive instructions from a user including charging amount requests and, in some examples, payment information. A controller **39** is provided in the charging base **12** and is configured to communicate with the electrical charging components **22**, the user interface panel **30**, the motor **34** and the vehicle charge connector **42**. It will be appreciated that the vehicle charge connector **42** is generically represented but can be configured to be plug types specific for use with any vehicle charging situation (e.g., Type 1, 110V; Type 2, 220V).

[0034] With additional reference now to FIGS. **2**A-**4**B, additional features of the charging cable coupling assembly **20** will be described. The charging cable coupling assembly **20** includes a charging cable **40** and a vehicle charge connector **42**. The vehicle charge connector **42** includes a charger body **50**, a charger handle portion **54** and a male extension portion **56**. The male extension portion **56** is configured to be received by a female receiving portion **66** (FIGS. **4**A-**4**B) provided on an electrified vehicle **70** (FIG. **3**).

[0035] An actuator **60** is movably disposed on the charger body **50** and configured to translate from a retracted or at-rest position (FIG. **2**A) to a deployed position (FIG. **2**B) upon receipt of a decoupling signal from the controller **39**. The actuator **60** can be configured to translate along a track assembly **62**. In examples, the track assembly **62** can be configured as posts however other configurations are contemplated. As identified above, translation of the actuator **60** from the retracted position shown in FIG. **2**A to the deployed position shown in FIG. **2**B causes the vehicle charge connector **42** to automatically detach from the vehicle **70**. As will be described below, the detaching is facilitated by engagement of a forcing surface **78** configured on a distal end of the actuator **60** with an engaging surface on the vehicle **70**.

[0036] Upon indication of a decoupling signal, the vehicle charge connector **42** automatically decouples from the vehicle **70**. As described herein, the decoupling signal that triggers the actuation of the actuator **60** from the at-rest position (FIG. **2**A) to the deployed position (FIG. **2**B) can result from a variety of conditions including an indication that charging has reached the desired limit, an unsafe condition such as a short circuit, unauthorized use of the charging system, a vehicle shift from park, and emergency use of a connected vehicle.

[0037] The electrified vehicle **70** generally includes an electrified powertrain **80** comprising one or more electric motors **82**. The electric motor(s) **82** are powered by a high voltage battery system **88** (e.g., a 16 kilowatt-hour (kWh) lithium-ion battery pack) and generate drive torque that is transferred to a driveline **90** of the vehicle **70**. The high-voltage battery system **88** according to the example shown is configured as a 48 volt battery system (1 kilowatt-hour) although other voltages are contemplated.

[0038] With particular reference now to FIGS. **3**-**4**B, an exemplary charging sequence will be described. At the outset, a user **100** can retrieve the vehicle charge connector **42** from the vehicle charging system **10**. In examples, the vehicle charge connector **42** can be magnetically coupled to the magnet **32** (see solid line location of the vehicle charge connector **42**). The user **100** can interact with the user interface panel **30** to enter the requested charge parameters (e.g., charge amount desired, time desired, or any parameter). It is further appreciated that in some examples where the vehicle charging system is in a public area, the user interface panel **30** can also receive payment information from the user **100**.

[0039] The user then moves the vehicle charge connector **42** off of the magnet **32** and locates the vehicle charge connector **42** into a vehicle charging port **110** (FIG. **4**A) provided on the vehicle **70**. In examples, the vehicle charging port **110** defines the female receiving portion **66**. In this regard, the male extension portion **56** is inserted by the user **100** into the female receiving portion **66** of the vehicle charging port **110** (see charging or connected position FIG. **4**A). In the charging position, the vehicle charge connector **42** is electrically and mechanically connected to the vehicle charging port **110** for transferring current from the vehicle charging system to the battery **88** of the vehicle **70**. Typically, during charging, the user **100** does not need to stand by the vehicle charge connector **42** and is generally free to return to the vehicle cabin or move away from the vehicle **70** entirely.

[0040] Once charging is complete (or other decoupling conditions described herein are met), the controller **39** sends a signal to the actuator **60** on the vehicle charge connector **42** causing the actuator **60** to move from the connected position (FIG. **4**A) to the decoupled position (FIG. **4**B). During movement of the actuator **60** from the connected position (FIG. **4**A) to the decoupled position (FIG. **4**B), the forcing surface **78** on the actuator **60** urges against an engagement surface **122** on the vehicle **70** resulting in the male insertion portion **56** to be forcibly retracted from the female receiving portion **66** and the vehicle charge connector **42** to move to the decoupled position (FIG. **4**B). In examples, this action can additionally cause the vehicle charge connector **42** to move generally in a direction **128** away from the vehicle **70**. It is appreciated that additional mechanical decoupling may also occur between the vehicle charge connector **42** and the vehicle charging port **110** during the movement of the actuator **60** to the decoupled position.

[0041] Once the actuator **60** moves to the decoupled position and the vehicle charge connector **42** is retracted from the vehicle charging port **110**, the controller **39** sends a signal to the motor **34** to rotate shaft **36** causing the arm **14** to retract toward the charging base **12** (see FIGS. **5** and **6**). The vehicle charge connector **42** is moved to a position proximate the magnet **32** such that the vehicle charge connector **42** is magnetically coupled to the magnet **32** (see solid line location of the vehicle charge connector **42**, FIG. **6**).

[0042] With reference to FIG. **7** a method of automatically decoupling the vehicle charge connector **42** from the electrified vehicle **70** is shown generally at reference **200**. Control starts at **210**. At **212** control determines whether a vehicle charge event has been enabled. If not, control loops to **212**. If control determines that a vehicle charge event has started at **212**, control determines whether a charge is complete at **220**. A complete charge is used herein to denote a charging sequence has reached a requested state by a user such as a desired partial charge amount or reached a full charge. If yes, control proceeds to **230** where charging ends. Charging ending can be used to denote a conclusion of electrical current being communicated from the vehicle charging system **10** to the vehicle **70**.

[0043] If control determines that charge is not complete at **220**, control determines whether an unsafe condition exists, such as a short circuit at **222**. If yes, control proceeds to **230**. If control determines that an unsafe condition does not exist at **222**, control determines whether the vehicle **70** has shifted out of park or other disconnect condition has been satisfied at **226**. If yes, control proceeds to **230**. If control determines that a shift or other disconnect condition has not been satisfied at **226**, control loops to **220**.

[0044] After charging has ended at **230**, control communicates, at **240**, a signal to the actuator **60** to move from the retracted position to the deployed position to unplug the vehicle charge connector **42** from the vehicle charging port **110**. At **244** control commands the arm **14** to retract toward the charging base **12** (from the location shown in FIG. **5** back to the location shown in FIG. **6**) where the vehicle charge connector **42** magnetically couples to the magnet **32**.

[0045] It will be understood that the mixing and matching of features, elements, methodologies, systems and/or functions between various examples may be expressly contemplated herein so that one skilled in the art will appreciate from the present teachings that features, elements, systems and/or functions of one example may be incorporated into another example as appropriate, unless described otherwise above. It will also be understood that the description, including disclosed examples and drawings, is merely exemplary in nature intended for purposes of illustration only and is not intended to limit the scope of the present disclosure, its application or uses. Thus, variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure.

Claims

1. A vehicle charging system for an electrified vehicle, the vehicle charging system comprising: a charging base that houses electrical charging components; a cable arm movably coupled to the charging base; a charging cable configured to communicate current from the electrical charging components to the electrified vehicle; a controller that is configured to determine whether a decoupling condition has been satisfied and communicate a decoupling signal in response thereto; and a vehicle charge connector extending from the charging cable and configured to be electrically coupled to a vehicle charging port of the electrified vehicle during a charging event, the vehicle charge connector comprising: a charger body having a male extension portion extending therefrom; and an actuator movably disposed on the charger body, the actuator configured to move from a retracted position to a deployed position upon receipt of the decoupling signal, wherein movement of the actuator to the deployed position causes a forcing surface on the actuator to push against a surface on the electrified vehicle and urge the vehicle charge connector away from the vehicle charging port.
2. The vehicle charging system of claim 1, further comprising: a motor disposed on the charging base and mechanically coupled to the cable arm, the motor configured to move the cable arm from an outward charging position to a retracted position.
3. The vehicle charging system of claim 2, wherein the controller communicates a signal to the motor to move the cable arm to the retracted position subsequent to communicating the decoupling signal.
4. The vehicle charging system of claim 1, wherein the charging base further comprises a magnet, wherein the magnet is configured to magnetically couple the vehicle charge connector thereat.
5. The vehicle charging system of claim 1, wherein the controller determines that a decoupling condition has been satisfied based on a vehicle charge is complete.
6. The vehicle charging system of claim 1, wherein the controller determines that a decoupling condition has been satisfied based on an unsafe condition being detected.
7. The vehicle charging system of claim 1, wherein the controller determines that a decoupling condition has been satisfied based on an indication that the electrified vehicle has shifted out of park.
8. The vehicle charging system of claim 1, wherein the charger body includes a track assembly, wherein the actuator translates along the track assembly from the retracted position to the deployed position.
9. The vehicle charging system of claim 1, wherein the vehicle charging port further defines a female receiving portion that receives the male extension portion of the charger body during the charging event.
10. A method for automatically decoupling a vehicle charge connector from an electrified vehicle, the method comprising: determining, at a controller provided in a charging base that houses electrical charging components, that a charge condition is enabled; determining, at the controller, whether a decoupling condition has been satisfied; and communicating, based on a decoupling condition being satisfied, a decoupling signal to an actuator on the vehicle charge connector, the actuator being caused to move from a retracted position to a deployed position upon receipt of the decoupling signal, wherein movement of the actuator to the deployed position causes a forcing surface on the actuator to push against a surface on the electrified vehicle and urge the vehicle charge connector away from the vehicle charging port.
11. The method of claim 10, further comprising: communicating a signal to a motor disposed in the charging base, the motor mechanically coupled to a cable arm, whereby the motor moves the cable arm from an outward charging position to a retracted position.
12. The method of claim 11, wherein the signal is communicated to the motor subsequent to

communicating the decoupling signal.
13. The method of claim 12, further comprising magnetically coupling the vehicle charge connector to a magnet disposed on the charging base subsequent to the cable arm moving to the retracted position.
14. The method of claim 10, wherein the decoupling condition is satisfied based on an indication that a vehicle charge is complete.
15. The method of claim 10, wherein the decoupling condition is satisfied based on an indication that an unsafe condition is detected.
16. The method of claim 10, wherein the decoupling condition is satisfied based on an indication that the electrified vehicle has shifted out of park.
17. The method of claim 10, wherein moving the actuator from the retracted position to the deployed position comprises translating the actuator along a track.